



Customer Shopping Behavior Analysis

3,900 purchases — 18 features — goal: actionable insights

Dataset Summary

Rows / Columns

3,900 rows · 18
columns

Key Features

Demographics,
Purchase details,
Behavior, Shipping

Missing Data

37 missing Review
Rating values



Exploratory Data Prep

01

Load & Inspect

`pandas df.info() · .describe()`

02

Clean & Impute

Imputed Review Rating by category median

03

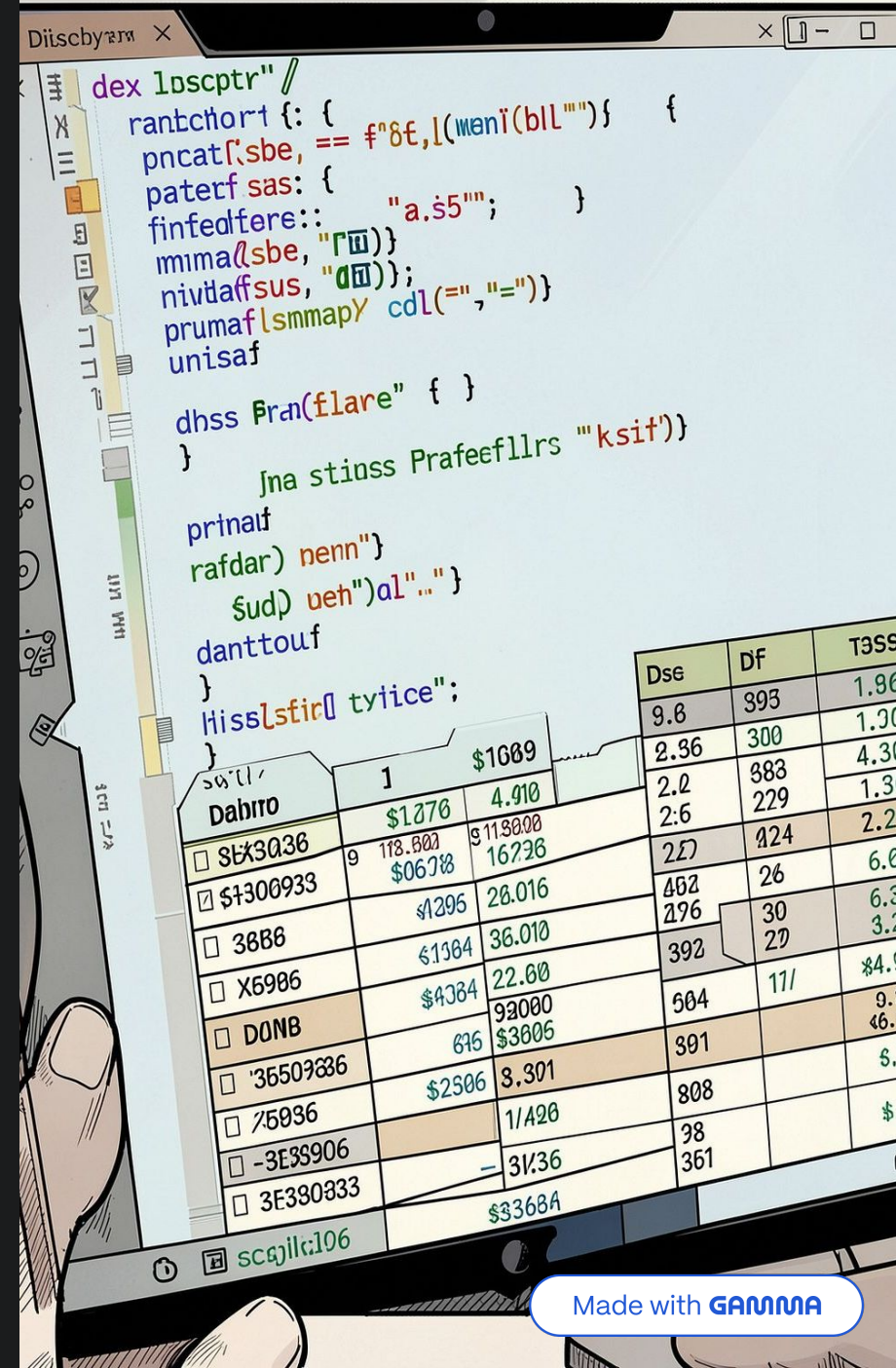
Standardize

Renamed columns to snake_case

04

Feature Eng.

`age_group · purchase_frequency_days`



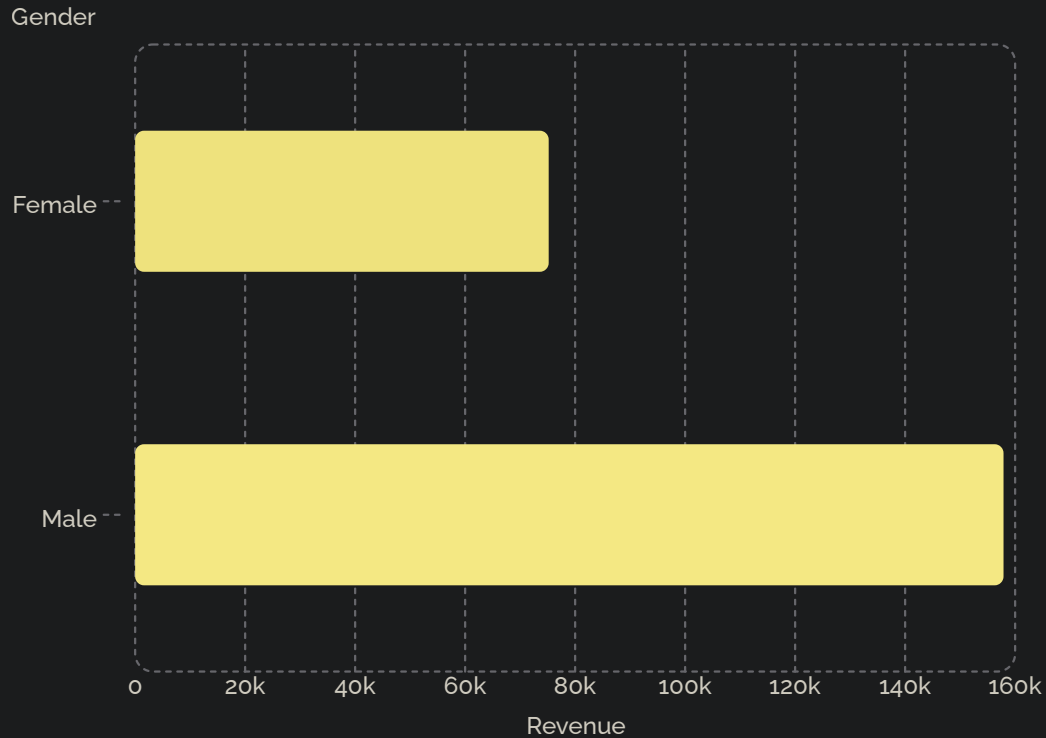
Database Integration



Workflow

- Cleaned DataFrame loaded to PostgreSQL
- SQL used for structured business queries
- Analysis reproducible and queryable

Key SQL Findings — Revenue & Spend



Avg Purchase

\$59.76

Avg Rating

3.75

Express vs Standard: \$60.48 vs \$58.46

Top Products & Discounts



Top Rated
Gloves — 3.86



Top Rated
Sandals — 3.84



Discount-
Dependent
Hat — 50%
discounted
purchases



Discount-
Dependent
Sneakers — 49.66%

Customer Segments



Loyal

3116 customers

Returning

701 customers

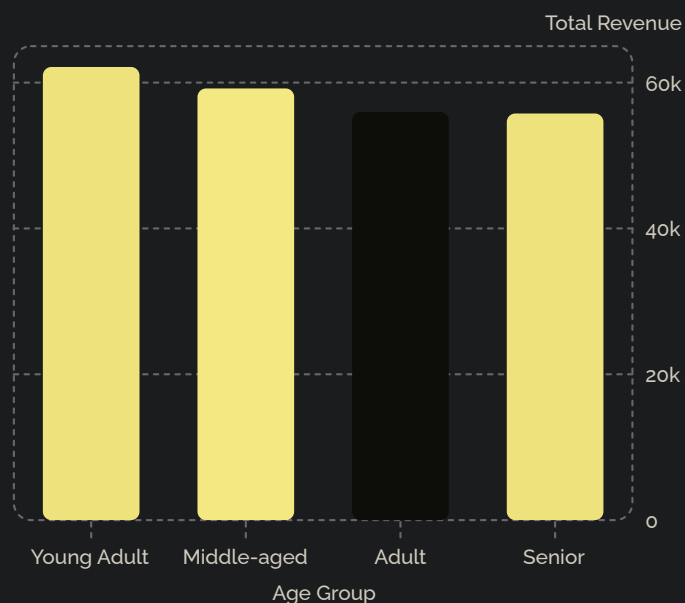
New

83 customers

Majority Loyal (3,116). Tail of New customers (83).

Age Groups & Revenue

Target high-revenue age groups: Young Adult · Middle-aged



Power BI Dashboard Highlights



- 3.9K customers · Avg purchase \$59.76 · Avg rating 3.75
- Subscription: 27% Yes / 73% No
- Revenue by Category: Clothing 100K · Accessories 75K · Footwear 30K · Outerwear 15K
- Sales by Category: Clothing 1,800 · Accessories 1,200 · Footwear 600 · Outerwear 200

Business Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits

Loyalty Programs

Reward repeat buyers to grow Loyal segment

Review Discount Policy

Balance sales lift with margin control

Product Positioning

Highlight top-rated & best-selling items

Targeted Marketing

Focus on high-revenue age groups & express-shipping users